

ZICHEN XIE

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🏠 [Homepage](#) ✧ 📄 [Google Scholar](#) ✧ 🐙 [GitHub Profile](#)

RESEARCH INTERESTS

My research centers on **code generation**, with the goal of improving the correctness and safety of LLM-generated code. I currently work on **verifiable code generation**: having models generate code together with a formal proof of its correctness that a machine can check automatically, so correctness is formally guaranteed rather than empirically tested. More broadly, I am interested in Software Engineering, Software Security, and Language Models.

- **Improving Code Generation Quality.** Enforcing functional-correctness and security constraints during diffusion-based code decoding [1], and benchmarking end-to-end verifiable code generation on competitive-programming tasks [2]. I also propose Verification Chain-of-Thought (VCoT), exposing fine-grained verification reasoning to evaluate LLMs and supervise future proof-generation training [3].
- **LLM for Program Analysis & Bug Detection.** Synthesizing specialized static-analysis checkers from historical bug patterns for scalable LLM-based detection [4], uncovering **92** long-latent Linux Kernel bugs (30 CVEs), and rethinking cryptographic API misuse detection with LLMs [5].

EDUCATION

- **University of Virginia, PhD Student** *Aug. 2025 - Expected May 2030*
Ph.D. Student in Computer Science, advised by [Prof. Wenxi Wang](#)
- **Zhejiang University** *Sept. 2021 - Jun. 2025*
B.Eng. in Information Security GPA: 3.97/4, 90.90/100
Member of ACEE ([Chu Kochen Honors College](#))

PUBLICATION

- [1] **Constrained Code Generation with Discrete Diffusion** [\[Paper\]](#)
arXiv Preprint 2026
Lize Shao, Michael Cardei, [Zichen Xie](#), Ferdinando Fioretto, Wenxi Wang
- [2] **VeriContest: A Competitive-Programming Benchmark for Verifiable Code Generation** [\[Paper\]](#) [\[Code\]](#)
arXiv Preprint 2026
[Zichen Xie](#), Mrigank Pawagi, Yuxin Liu, Aaditi Rai, Lize Shao, John Berberian Jr., Sicong Che, Wenxi Wang
- [3] **Can LLMs Reason Like Automated Theorem Provers for Rust Verification? VCoT-Bench: Evaluating via Verification Chain of Thought** [\[Paper\]](#) [\[Code\]](#)
ICML 2026. The 43rd International Conference on Machine Learning
[Zichen Xie](#), Wenxi Wang
- [4] **KNighter: Transforming Static Analysis with LLM-Synthesized Checkers** [\[Paper\]](#) [\[Code\]](#)
SOSP 2025. The 31st Symposium on Operating Systems Principles
Chenyuan Yang, Zijie Zhao, [Zichen Xie](#), Haoyu Li, Lingming Zhang
- [5] **Beyond Static Pattern Matching? Rethinking Automatic Cryptographic API Misuse Detection in the Era of LLMs** [\[Paper\]](#)
ISSTA 2025. The 34th ACM SIGSOFT International Symposium on Software Testing and Analysis
Yifan Xia, [Zichen Xie](#), Peiyu Liu, Kangjie Lu, Yan Liu, Wenhai Wang, Shouling Ji

PROFESSIONAL EXPERIENCE

- **Research Assistant at Hiprel Group, UVA** *Aug. 2025 - Current*
VA, US
Advised by [Prof. Wenxi Wang](#)
Topic: verifiable code generation
- **Software Engineer Intern at NetEase Games, ThunderFire** *Apr. 2025 - May 2025*
Hangzhou, China
Hosted by Yuanyuyang Liu
Topic: game testing
- **Software Engineer Intern at Tencent CDG** *Jul. 2024 - Sept. 2024*
Shenzhen, China
Hosted by Na Li and Bingxiang Xu
Topic: MLLMs for advertisement testing
- **Research Intern at PL/FM/SE Group, UIUC** *May 2024 - Mar. 2025*
IL, US
Advised by [Prof. Lingming Zhang](#)
Topic: LLMs for static analysis
- **Research Assistant at NESA Lab, Zhejiang University** *Dec. 2023 - Jun. 2024*
Hangzhou, China
Advised by [Prof. Shouling Ji](#)
Topic: LLMs for cryptographic API misuse detection

AWARDS AND HONORS

- **Zhejiang Provincial Government Scholarship** *Nov. 2024*
- **National Second Prize of the China Undergraduate Mathematical Contest in Model** *Oct. 2023*
- **Third-Class Scholarship for Outstanding Students** *Oct. 2023*
- **Zhejiang Provincial Government Scholarship** *Oct. 2022*

SKILL STACK

- **Languages:** Python, C/C++, Java, Rust, SQL.
- **Tools & Frameworks:** Git, L^AT_EX, Verus, Z3, PyTorch.